

and 650 pixels from the left of the monitor screen area. To have a continuous animation of the plot, the variables are incremented as and when they get executed by the afore-mentioned equations. Since JavaScript program counts time in milliseconds, the plot refreshes every one millisecond using the code 'window.setTimeout('Lissajous()', 1).'

The program was successfully tested on a 2GB DDR3 random-access memory (RAM). If your PC has lower

RAM, increase the milliseconds figure '1' to '250', '500' or '1000.'

This application requires Internet Explorer and JavaScript plug-in, which has to be initialised or installed in the system. If you have the plug-in and it is disabled, a light yellow bar with warning message appears on the screen when the program is run (refer Screenshot 2). Right-click the yellow bar and choose 'Allow Blocked Content' option fol-

lowed by clicking 'Yes' button from the dialogue box. The application should start running. Else, install the plug-in from the Internet or a reliable disk.

EFY note. The source code of this article is available on www.efymag.com website.

The author is a B.Tech in electronics and communications engineering, likes coding JavaScript, and has keen interest in space technology (satellite communications) and mobile communication

LISSAJOUS.HTM

```
<html>
<head>
<title>JavaScript based Lissajous Curve
Simulator Interface</title>
<meta http-equiv="Page-Enter" content="true"
evalTrans(Duration=1.0,Transition=23)">
<meta http-equiv="Page-Exit" content="true"
blendTrans(Duration=1.0)">
<meta http-equiv="Page-Exit" content="true"
evalTrans(Duration=1.0,Transition=23)">
</style>
.L{border:1px solid black;font-family:Palatino;
font-size:16px;color:red;
background-color:black;text-align:center;overflow:auto;}
</style>
<script class=L>
This Application Requires JavaScript Plug-in which has to be initialized or installed.
It seems that your OS does not have the Plug-in Installed or has been disabled by you. Please Enable/Install the Plug-in.
If you have the plug-in and is currently disabled, right-click on the light-
```

```
yellow bar above, Choose "Allow Blocked Content..." and then click "Yes" from the consecutive dialog box, you should see the Application running, or else install the plug-in from Internet/Reliable Disc.</noscript>
</head>
<body oncontextmenu="return false" onselectstart="return false" oncopy="return false" ondragstart="return false" onfocus="return false" onblur="return false" onhelp="return false" class=L>
<div id=p class=L></div>
<script language=JavaScript>
a=10;
b=20;
d=1;
for(i=0;i<3600;i++)
{
document.write('<div id="Sc" style="position:absolute;color:#cda861;">.</div>');
function Lissajous()
```

```
{
for(i=0;i<3600;i++)
{
Sc[i].style.top=450+320*Math.sin(d*i*Math.PI/a);
Sc[i].style.left=650+320*Math.sin(d*i*Math.PI/b);
}
a++;
b++;
d++;
p.innerHTML+=JavaScript based Lissajous Curve Simulator Interface<br>Created by EFY Enterprises Pvt Ltd<br>Parameters of this curve: Channel 1 (X): "a+", Channel 2 (Y): "b", window.setTimeout('Lissajous()', 1);
}
window.onload=Lissajous;
</script>
</body>
</html>
```



Universal Robotics Development Platform

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